基于 MATLAB App Designer 的交互式图像分析系统设计

——代码备份

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我设计并实现了一个基于 *MATLAB App Designer* 的交互式图像处理与特征分析 *App*。本 *App* 实现了图像增强、*Canny* 边缘提取，以及图像灰度 *3D* 拓扑图的实时渲染。

```
classdef app1 < matlab.apps.AppBase

    % Properties that correspond to app components
    properties (Access = public)
        UIFigure           matlab.ui.Figure
        TabGroup           matlab.ui.container.TabGroup
        Tab                matlab.ui.container.Tab
        GridLayout         matlab.ui.container.GridLayout
        Panel              matlab.ui.container.Panel
        Btn_ExportData     matlab.ui.control.Button
        Btn_Reset          matlab.ui.control.Button
        Slider_Intensity   matlab.ui.control.Slider
        Label_2            matlab.ui.control.Label
        Drop_Algo          matlab.ui.control.DropDown
        Label              matlab.ui.control.Label
        Btn_Load           matlab.ui.control.Button
        UIAxes_Result      matlab.ui.control.UIAxes
        UIAxes_Original    matlab.ui.control.UIAxes
        Tab_2              matlab.ui.container.Tab
        GridLayout3        matlab.ui.container.GridLayout
        Btn_ROI            matlab.ui.control.Button
        UIAxes_SegMain     matlab.ui.control.UIAxes
        UIAxes_SegCrop     matlab.ui.control.UIAxes
        DTab               matlab.ui.container.Tab
        GridLayout2        matlab.ui.container.GridLayout
        Btn_3D             matlab.ui.control.Button
        UIAxes_3D          matlab.ui.control.UIAxes
    end


    properties (Access = private)
    ImgOriginal
    ImgCurrent
        Btn_Export         matlab.ui.control.Button
        Btn_Export3D       matlab.ui.control.Button
end

    methods (Access = private)

        function processAndShowCrop(app, currentPosition)

            croppedImg = imcrop(app.ImgOriginal, currentPosition);

            if size(croppedImg, 3) == 3
                grayCrop = rgb2gray(croppedImg);
            else
```

```
                grayCrop = croppedImg;
            end


            edgeCrop = edge(grayCrop, 'Canny');


            imshow(edgeCrop, 'Parent', app.UIAxes_SegCrop);
            title(app.UIAxes_SegCrop, '鬈鬈氈氈橡鬈橡
                                    ');endendmethods (Access = private)function
startupFcn(app)endfunction BtnLoadButtonPushed(app, event)[file, path] =
uigetfile('*.jpg;*.png;*.bmp;*.tif','');if isequal(file, 0)return; endfullFileName =
fullfile(path, file); app.ImgOriginal = imread(fullFileName); app.ImgCurrent =
app.ImgOriginal; imshow(app.ImgOriginal,'Parent', app.UIAxesOriginal); imshow(app.ImgCurrent,'Parent', app.UIAx
app.DropAlgo.Value; d = uiprogressdlg(app.UIFigure,'Title','L−','Message',['c(gL :'
, selectedAlgo]); tryswitchselectedAlgocase'P6:'clipLimitValue = 0.1; if size(app.ImgOriginal, 3) ==
3labImg = rgb2lab(app.ImgOriginal); L = labImg(:, :, 1)/100; Lenhanced =
adapthisteq(L,'NumTiles', [44],'ClipLimit', clipLimitValue); labImg(:, :, 1) =
Lenhanced * 100; app.ImgCurrent = lab2rgb(labImg); elseapp.ImgCurrent =
adapthisteq(app.ImgOriginal,'NumTiles', [44],'ClipLimit', clipLimitValue); endcase'*irΞ'if size(app.ImgOriginal, 3) ==
3grayImg = rgb2gray(app.ImgOriginal); elsegrayImg = app.ImgOriginal; endapp.ImgCurrent =
ind2rgb(grayImg, jet(256)); case'¸'if size(app.ImgOriginal, 3) == 3grayImg =
rgb2gray(app.ImgOriginal); elsegrayImg = app.ImgOriginal; endedgeImg =
edge(grayImg,'Canny'); app.ImgCurrent = uint8(edgeImg) *
255; endimshow(app.ImgCurrent,'Parent', app.UIAxesResult); catchMEuialert(app.UIFigure, ['ΞΣ1% :'
ME.message],'ß'); endclose(d); value =
app.DropAlgo.Value; endfunctionBtnResetButtonPushed(app, event)app.ImgCurrent =
app.ImgOriginal; imshow(app.ImgCurrent,'Parent', app.UIAxesResult); endfunctionBtn3DButtonPushed(app, event)if
uiprogressdlg(app.UIFigure,'Title',' æ5','Message', c(L3D2fiflß...'); if size(app.ImgOriginal, 3) ==
3grayImg = rgb2gray(app.ImgOriginal); elsegrayImg = app.ImgOriginal; endsmallImg =
imresize(grayImg, [200, 200]); surf(app.UIAxes3D, double(smallImg),'EdgeColor','none'); colormap(app.UIAxes3D,'paru
drawrectangle(app.UIAxesSegMain,'Label','ϒ:','Color','r','LineWidth', 2); app.processAndShowCrop(roi.Position); addl
app.SliderIntensity.Value; if isempty(app.ImgOriginal)return; endsliderVal =
app.SliderIntensity.Value; selectedAlgo =
app.DropAlgo.Value; switchselectedAlgocase'P6:'clipLimitValue =
0.01 + (sliderVal/100) * 0.19; if size(app.ImgOriginal, 3) == 3labImg = rgb2lab(app.ImgOriginal); L =
labImg(:, :, 1)/100; Lenhanced = adapthisteq(L,'NumTiles', [44],'ClipLimit', clipLimitValue); labImg(:, :
, 1) = Lenhanced * 100; app.ImgCurrent = lab2rgb(labImg); elseapp.ImgCurrent =
adapthisteq(app.ImgOriginal,'NumTiles', [44],'ClipLimit', clipLimitValue); endcase'¸'thresh =
0.1 + ((100 − sliderVal)/100) * 0.7; if size(app.ImgOriginal, 3) == 3grayImg =
rgb2gray(app.ImgOriginal); elsegrayImg = app.ImgOriginal; endedgeImg =
edge(grayImg,'Canny', thresh); app.ImgCurrent = uint8(edgeImg) * 255; case'*irΞ'alpha =
sliderVal/100; if size(app.ImgOriginal, 3) == 3grayImg = rgb2gray(app.ImgOriginal); elsegrayImg =
app.ImgOriginal; endpseudoImg = ind2rgb(grayImg, jet(256)); grayRGB =
cat(3, grayImg, grayImg, grayImg)/255; app.ImgCurrent = alpha * pseudoImg + (1 − alpha) *
double(grayRGB); endimshow(app.ImgCurrent,'Parent', app.UIAxesResult); endfunctionBtnExportDataPushed(app, eve
uiputfile(...'*.png','_PNG(*.png)';...'*.jpg','JPEG(*.jpg)';...'*.bmp','MBMP(*.bmp)',...'XΞΣ','ProcessedImage.png'
fullfile(path, file); tryd =
uiprogressdlg(app.UIFigure,'Title','−','Message','c(e...'); imwrite(app.ImgCurrent, fullFileName); close(d); uialert(app
, ME.message],'1%','Icon','error'); endendendmethods(Access =
private)functioncreateComponents(app)app.UIFigure =
uifigure('Visible','off'); app.UIFigure.Color = [000]; app.UIFigure.Position =
[100100624456]; app.UIFigure.Name ='MATLABApp'; app.UIFigure.WindowState ='
maximized'; app.TabGroup = uitabgroup(app.UIFigure); app.TabGroup.Position =
[1 − 23640480]; app.Tab = uitab(app.TabGroup); app.Tab.Title ='z:'; app.GridLayout =
uigridlayout(app.Tab); app.GridLayout.ColumnWidth =
'1.45x','1x','1.36x','1.44x'; app.GridLayout.RowHeight = 177,'1x'; app.GridLayout.ColumnSpacing =
6; app.GridLayout.Padding = [610610]; app.UIAxesOriginal =
uiaxes(app.GridLayout); title(app.UIAxesOriginal,''); xlabel(app.UIAxesOriginal,'X'); ylabel(app.UIAxesOriginal,'Y'); zlab
TimesNewRoman'; app.UIAxesOriginal.FontWeight ='bold'; app.UIAxesOriginal.XColor =
[0.80.80.8]; app.UIAxesOriginal.YColor = [0.80.80.8]; app.UIAxesOriginal.Layout.Row =
2; app.UIAxesOriginal.Layout.Column = [12]; app.UIAxesResult =
uiaxes(app.GridLayout); title(app.UIAxesResult,'ΞΣ'); xlabel(app.UIAxesResult,'X'); ylabel(app.UIAxesResult,'Y'); zlabel(a
TimesNewRoman'; app.UIAxesResult.FontWeight ='bold'; app.UIAxesResult.XColor =
[0.6510.6510.651]; app.UIAxesResult.YColor = [0.6510.6510.651]; app.UIAxesResult.Layout.Row =
2; app.UIAxesResult.Layout.Column = [34]; app.Panel = uipanel(app.GridLayout); app.Panel.Title ='
6'; app.Panel.Layout.Row = 1; app.Panel.Layout.Column = [14]; app.BtnLoad =
uibutton(app.Panel,'push'); app.BtnLoad.ButtonPushedFcn =
createCallbackFcn(app, @BtnLoadButtonPushed, true); app.BtnLoad.Position =
[3812010022]; app.BtnLoad.Text ='e'; app.Label = uilabel(app.Panel); app.Label.HorizontalAlignment ='
right'; app.Label.Position = [38765322]; app.Label.Text ='Ψ'; app.DropAlgo =
```

$uidropdown(app.Panel); app.Drop_Algo.Items = {}'P6:',' *ir\Xi',' ¸'}; app.Drop_Algo.ValueChangedFcn =$
$createCallbackFcn(app, @Drop_AlgoValueChanged, true); app.Drop_Algo.Position =$
$[1067610022]; app.Drop_Algo.Value =' P6:'; app.Label_2 =$
$uilabel(app.Panel); app.Label_2.HorizontalAlignment =' right'; app.Label_2.Position =$
$[39365322]; app.Label_2.Text =' ::'; app.Slider_Intensity =$
$uislider(app.Panel); app.Slider_Intensity.ValueChangedFcn =$
$createCallbackFcn(app, @Slider_IntensityValueChanged, true); app.Slider_Intensity.Position =$
$[114451503]; app.Btn_Reset = uibutton(app.Panel,' push'); app.Btn_Reset.ButtonPushedFcn =$
$createCallbackFcn(app, @Btn_ResetButtonPushed, true); app.Btn_Reset.Position =$
$[26112110022]; app.Btn_Reset.Text =' bfl'; app.Btn_ExportData =$
$uibutton(app.Panel,' push'); app.Btn_ExportData.ButtonPushedFcn =$
$createCallbackFcn(app, @Btn_ExportDataPushed, true); app.Btn_ExportData.Position =$
$[15112110022]; app.Btn_ExportData.Text =' '; app.Tab_2 = uitab(app.TabGroup); app.Tab_2.Title ='$
$*'; app.GridLayout3 = uigridlayout(app.Tab_2); app.GridLayout3.ColumnWidth =$
$'1x', 46, 45,' 1.04x'; app.GridLayout3.RowHeight =$
$'1x', 22,' 1.14x',' 11.28x',' 8.62x'; app.GridLayout3.ColumnSpacing =$
$4.86001586914062; app.GridLayout3.Padding =$
$[4.8600158691406210.8600158691406210]; app.UIAxes_SegCrop =$
$uiaxes(app.GridLayout3); title(app.UIAxes_SegCrop,' H')xlabel(app.UIAxes_SegCrop,' X')ylabel(app.UIAxes_SegCrop,' Y')z$
$InkFree'; app.UIAxes_SegCrop.FontWeight =' bold'; app.UIAxes_SegCrop.XColor =$
$[0.6510.6510.651]; app.UIAxes_SegCrop.YColor = [0.6510.6510.651]; app.UIAxes_SegCrop.Layout.Row =$
$4; app.UIAxes_SegCrop.Layout.Column = [34]; app.UIAxes_SegMain =$
$uiaxes(app.GridLayout3); title(app.UIAxes_SegMain,' ')xlabel(app.UIAxes_SegMain,' X')ylabel(app.UIAxes_SegMain,' Y')z$
$InkFree'; app.UIAxes_SegMain.FontWeight =' bold'; app.UIAxes_SegMain.XColor =$
$[0.6510.6510.651]; app.UIAxes_SegMain.YColor = [0.6510.6510.651]; app.UIAxes_SegMain.ZColor =$
$[0.6510.6510.651]; app.UIAxes_SegMain.Layout.Row = 4; app.UIAxes_SegMain.Layout.Column =$
$[12]; colormap(app.UIAxes_SegMain,' winter')app.Btn_ROI =$
$uibutton(app.GridLayout3,' push'); app.Btn_ROI.ButtonPushedFcn =$
$createCallbackFcn(app, @Btn_ROIButtonPushed, true); app.Btn_ROI.Layout.Row =$
$2; app.Btn_ROI.Layout.Column = [23]; app.Btn_ROI.Text =' *'; app.DTab =$
$uitab(app.TabGroup); app.DTab.Title =' 3D\Xi0'; app.GridLayout2 =$
$uigridlayout(app.DTab); app.GridLayout2.ColumnWidth =$
$'1.03x', 100,' 1x'; app.GridLayout2.RowHeight = 56,' 1x',' 16.1x'; app.GridLayout2.ColumnSpacing =$
$2.5; app.GridLayout2.Padding = [2.5102.510]; app.UIAxes_3D =$
$uiaxes(app.GridLayout2); title(app.UIAxes_3D,' 3D0')xlabel(app.UIAxes_3D,' X')ylabel(app.UIAxes_3D,' Y')zlabel(app.UIA$
$3; app.UIAxes_3D.Layout.Column = [13]; app.Btn_3D =$
$uibutton(app.GridLayout2,' push'); app.Btn_3D.ButtonPushedFcn =$
$createCallbackFcn(app, @Btn_3DButtonPushed, true); app.Btn_3D.Layout.Row =$
$1; app.Btn_3D.Layout.Column = 2; app.Btn_3D.Text =' Ø\i3D0'; app.UIFigure.Visible ='$
$on'; endendmethods(Access = public)functionapp =$
$app1createComponents(app)registerApp(app, app.UIFigure)runStartupFcn(app, @startupFcn)ifnargout ==$
$0clearappendendfunctiondelete(app)delete(app.UIFigure)endendend$